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RESEARCH ARTICLE

Characteristics of flow from an oxy-fuel burner with separated jets: influence of jet injection angle

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Abstract This paper presents an experimental study of flow development and structure on a separated jet burner in reacting and non-reacting flows. Effects of deflection jets in an aligned configuration of three round jets are emphasized. The idea is based on the confinement of a central jet of fuel by two side jets of oxygen to improve mixing, to control flame stability, and to reduce pollutant emissions. The fields of mean velocity and fluctuation intensity were measured using Particle Image Velocimetry. The deflection of jets has a considerable effect on the dynamic behavior and on the flame characteristics. Results showed that the deflection of jets favors mixing and accelerates merging and combining of jets to a single one. Measurements in reacting flow showed a high influence of combustion on dynamic fields. Compared to non-reactive case, in combustion, larger radial expansion and higher velocity were observed, particularly, above the stabilization point of the flame.

1 Introduction

Growing environmental concerns and the optimization of performance of industrial furnaces require the development of new generations of burners and improved combustion technologies. Manufacturers are turning to a new type of burners where fuel and oxidizer are injected separately. Design of these burners requires the knowledge of mechanisms controlling the stabilization of flame and the production of pollutants such as nitrogen oxides. The present paper reports the results of an experimental investigation of flow issued from burner designed by three aligned separated jets. The mixing and the dynamic field for both reacting and non-reacting flows are studied.

Multiple jets are used in wide variety of engineering applications, for example, thrust augmenting ejectors for VTOL/STOL aircraft, jet engine/rocket combustors, and industrial gas burners. It is well known that the multiple jets have many advantages over single jet, such as better mixing and noise reduction (Raghunathan and Reid 1981). Many investigations have been reported in the literature regarding the structure and development of non-reacting multiple jets (Tanaka 1974; Tanaka and Nakata 1975; Krothapalli et al. 1980; Pani and Dash 1983; Simonich 1986; Yimer et al. 1996; Moawad Ahmed et al. 2001). On the other hand, few studies of reacting flow in multiple jet configurations are conducted, let alone in oxy-fuel combustion. Menon and Gollahalli (1988) have studied the interaction of multiple jet flames of propane, resulting from 2, 3, and 5 nozzles subjected to a cross flow. Leite et al. (1996) were interested in the influence of jet number, the spacing between jets, and tube diameters on the length of an acetylene flame. Lenze et al. (1975) have studied the mutual influence of three and five jet diffusion flame, with town gas and natural gas burners. Their measurements concern concentrations, flame length, and flame width in free and confined multiple flames.

Combustion with oxygen is characterized by a higher adiabatic flame temperature, a higher flame velocity, a lower ignition temperature, and a wider flammability range than is the case in combustion with air (Baukal and Gebhart 1997; GEFGN 1979; Perthuis 1983; Ivernel and Marque 1975). Oxy-fuel burners have been adopted in a wide range of industrial furnaces to improve productivity and fuel

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efficiency, to reduce emissions of pollutants, and, in some applications, to improve product quality and yield, or to eliminate the capital and maintenance costs of air preheaters. Significant fractions of glass melting furnaces, steel scrap melting furnaces, electric arc furnaces, soaking pits, forging furnaces, ladles, aluminum melting furnaces, copper smelting and anode furnaces, hazardous waste incinerators, and rotary lead melting furnaces are fired with oxy-fuel burners (Baukal 2003).

Use of oxy-fuel burners with separated jets opens interesting possibilities in NOx reduction and modularity of flame properties (lift-off, topology, flame length) (Sautet et al. 2006; Boushaki et al. 2007). For separated jet burners, the principle is based on the geometrical separation of its nozzles. This design makes it possible to have a high dilution of reactants by combustion products, a large flame, and a homogeneous temperature in the flame. This configuration is highly three-dimensional and asymmetrical. Various parameters govern the flow from this type of burner like the number of jets, the form of nozzles, the spacing between the jets, the exit velocities, etc. In previous papers by the authors (Boushaki et al. 2007), the characteristics of flames in burners with separated jets are studied with burner parameters such as exit velocities, separation distance between jets and angle of injection. It is interesting to note that the inclination of jets makes it possible to have a better stability of flame and above all a significant reduction in NOx (Boushaki et al. 2008). The present paper considers the dynamic behavior of three jet interactions in more detail by varying angle of the side oxygen jets.

This study investigates the dynamic field on a burner with 25 kW power composed of three jets, one central jet of natural gas and two side jets of pure oxygen. The parameter of control consists in inclining the side oxygen jets toward the central jet of natural gas. The velocity measurements were carried out using particle image velocimetry (PIV) in non-reacting flow as well as in reacting flow inside the combustion chamber.

2 Experimental method

The experimental facility comprises a model furnace, which is 100-cm-high with square cross section (60 cm × 60 cm) and an oxy-fuel burner located on the bottom wall of the furnace (Fig. 1). The burner is made of three aligned tubes, one central tube of fuel and two side tubes of pure oxygen. The fuel used in this work is natural gas ($\rho_{ng} = 0.83\,\mathrm{kg/m^3}$), which molar composition is 85% CH_4; 9% C_2H_6; 3% C_3H_8; 2%N_2; 1% CO_2, and its net calorific value (NCV) is 45 MJ/kg. For safety reasons, when measurements are taken in non-reacting flow, the natural gas is

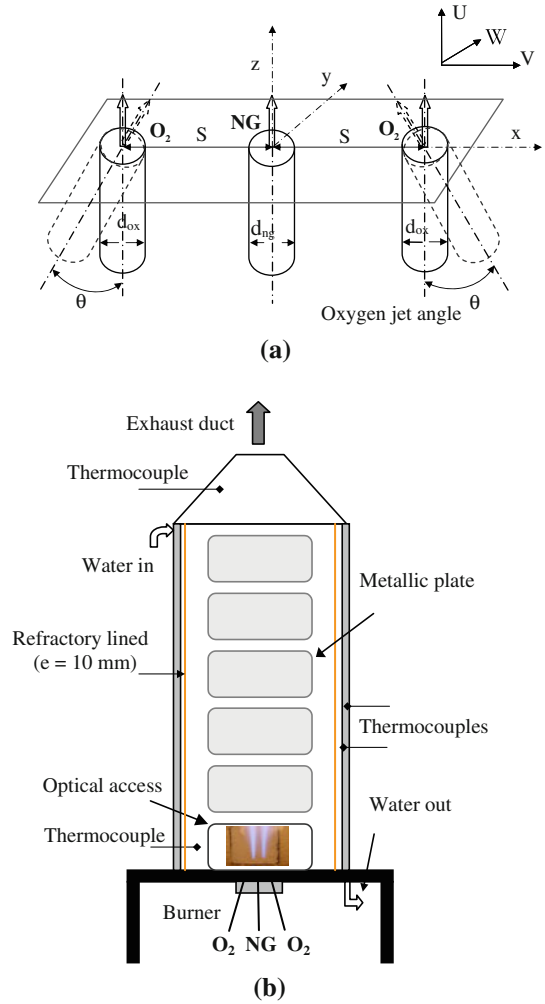

Fig. 1 Schematic view of the facility, **a** separated jet burner, **b** furnace

replaced by a mixture of 65.38% (in volume) of nitrogen and 34.62% of helium, a mixture with the same density as the used natural gas. However, the designation "natural gas: ng" is retained to discuss the central jet. Fuel and oxidizer flow rates are constant for all experiments to ensure constant power flames of 25 kW ($\dot{M}_{ng} = 556 \times 10^{-3}\,\mathrm{kg/s}$, $\dot{M}_{ox} = 1964 \times 10^{-3}\,\mathrm{kg/s}$). The flow rate of natural gas is controlled by a regulator of mass flow rate TYLAN RDM 280; the other gases (O_2, N_2 and He) are regulated by sonic throats calibrated by a flowmeter in function of pressure. The internal diameters of natural gas and oxygen tubes (d_{ng} and d_{ox}) are both 6 mm, their lengths are about 50 mm for each one. Exit velocities for the natural gas and the oxygen jets without control ($\theta = 0°$) inferred from the flow rates are, respectively, $U^0_{ng} = 23.7\,\mathrm{m/s}$ and $U^0_{ox} = 25.7\,\mathrm{m/s}$. The separation distance between the jets (S) is fixed at 12 mm. The control technique consists in inclining the side oxygen jets toward the natural gas jet as shown in Fig. 1. The angle of oxygen jets

(θ) compared to the vertical direction varies from 0° to 30°. In the case of reacting flow, the oxy-fuel flames develop in a confinement that models the industrial furnace. The lateral walls are refractory lined on the inside and water cooled on the outside. Optical access is provided through quartz windows in various places of the combustion chamber. The exhaust section is ended by a convergent with small opening (12 cm × 12 cm) to allow exhaust of the flue gases and access of a gas-sampling probe.

Experimental device for PIV measurement requires the basic elements used in laser tomography, i.e., a laser sheet that illuminates the zone of flow studied, a CCD camera, a PC for data acquisition, and a control unit for synchronization (Fig. 2). The laser used is double-pulsed Nd-YAG (Big Sky CFR200 Quantel) with a wavelength of 532 nm and a frequency of 10 Hz. Laser energy is adjustable and can be increased up to 150 mJ per pulse with pulse duration of 8 ns. The laser sheet is formed by a first divergent cylinder lens, which spreads out the beam then by second convergent spherical lens, which focuses the sheet (500 µm of thickness in the test section). The signal of Mie scattering emitted by particles is collected perpendicularly using a CCD camera FlowMaster of Lavision (12-bit dynamic and 1280 × 1024 pixels resolution) with a 50-mm lens F/1.2 Nikkon. An interferential filter at 532 nm is placed in front of the camera lens to reject visible light at wavelengths other than 532 nm. Time delay between the laser pulses was fixed at 10 µs. For each operation condition, up to 400 pairs of instantaneous images were collected.

The seeding particles for PIV measurements are olive oil particles (~3–4 µm in diameter) in non-reacting flow and zirconium oxide (ZrO_2) particles (~0.5 µm in diameter) for reacting flow. For olive oil particles, the seeding system is an atomizer based on Venturi principle to produce fine particles (Fig. 3a). It allows distribution of particles with more or less homogeneous sizes and in sufficient amount. In the reacting flow, the injection of ZrO_2 particles is performed using a seeding system made in our laboratory. The system consists of tubes equipped with porous plates, placed at the bottom at the level of exit gases (Fig. 3b). The gas passes through the porous medium and drags a certain quantity of particles providing a uniform seeding. Concentration of particles is controlled by valves through the gas flow rate in the line of seeding. The particles must be dried first, in order to limit agglomerates. Particles can modify the characteristics of the flow if they are introduced in high quantity and can even blow out the flame. Hence, the measurements were taken with seeding rates relatively low in order not to perturb the flow. The criterion assuring a good track of flow by particles is respected here, since the Stokes number obtained in the case of our experiments is much lower than unity (St $\ll$ 1). The Stokes number is defined as the ratio between the response time of particles (t_p) and a time characteristic of the flow (t_f). For example, $t_p \approx 0.015$ ms in the case of oxygen and oxide zirconium particles ($t_p = 2\rho_p r_p^2/9\mu_{CH_4}$: $\rho_p = 5600\,\mathrm{kg/m^3}$, $r_p = 0.5\,\mu\mathrm{m}$, $\mu_{CH_4} = 20.18 \times 10^{-6}$ Pa s), with the frequency of 1000 Hz, the Stokes number is about 0.015.

The software used for data acquisition and correlation processing was Davis of Lavision (DaVis 6.2.2). The size of images is 1280 × 1024 pixels that corresponds to 80 mm × 63 mm in physical size. The interrogation window size of PIV calculation is 32 × 32 pixels that corresponds to 1.97 mm × 1.97 mm^2 in physical size. The overlap is 75%, i.e., one vector per eight pixels (or 0.49 mm). Velocity vectors were determined using cross-correlation algorithm. The subpixel displacement was estimated by means of Gaussian peak of fitting. With a maximum displacement of eight pixels, this would correspond to less than 2% uncertainty in final velocity measurement. It was necessary to carry out a post processing to detect and correct the aberrant vectors which appear after cross-correlation calculations. Detection of false vectors can be done by the size of the vector. In this case, it is necessary to locate all vectors above a certain threshold of

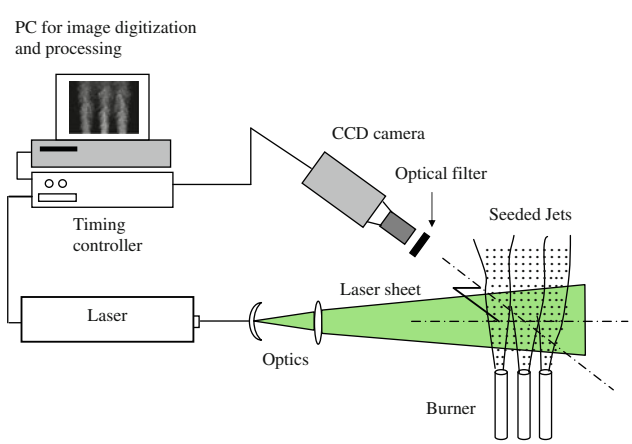

Fig. 2 Schematic diagram of the PIV setup (the CCD camera is perpendicular to the laser sheet)

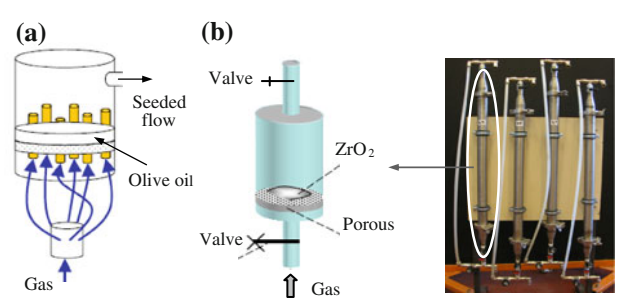

Fig. 3 Seeding systems, **a** for olive oil particles in non-reacting flow, **b** for ZrO_2 particles in reacting flow

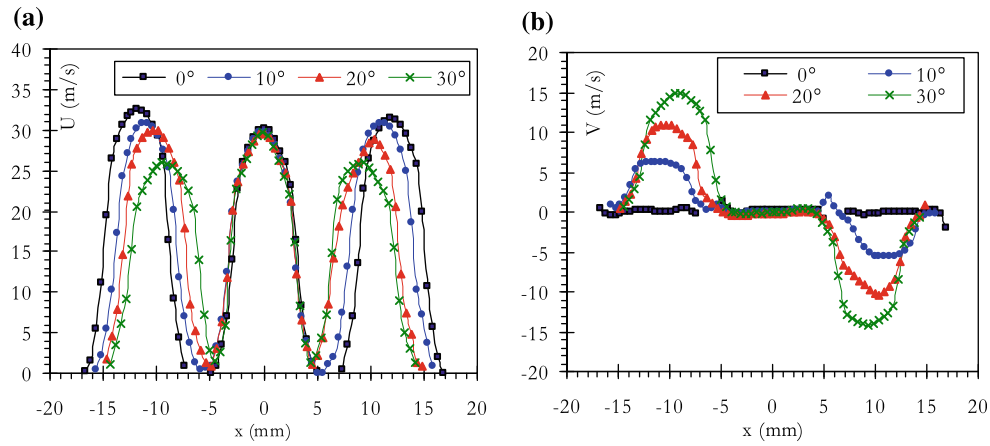

Fig. 4 Radial profiles of longitudinal (**a**) and transversal (**b**) velocities at exit nozzles ($z = 3$ mm) in non-reacting flow (oxygen jet angle, $\theta = 0°$, 10°, 20°, and 30°) [The error in velocity values is about 4% in average]

velocity according to the expected results. The direction of vectors can also help to identify false vectors knowing a priori the direction of flow. For that, the *allowable vector range* restricting the filtered vectors to a user specified in units of pixel was performed. A range may be specified for each component of velocity Vx, Vy, and Vz (range of ± value). Any vectors outside this range are removed. In some cases, in particular for instantaneous images, filters to refine the results are used as local median filter or regional median filter based on the neighboring vectors.

3 Results and discussion

3.1 Initial conditions of velocities

When injectors are inclined, the shape of nozzles changes becoming elliptical rather than round. Therefore, the profile of velocity for inclined jet at the exit nozzle is quite different to the one of a straight jet. That is why results of the exit velocities are provided; they are very useful for numerical studies. Figure 4 shows the profiles of mean velocities near the burner exit ($z = 3$ mm) in the non-reacting flow. For the side jets, the longitudinal mean velocity, U, decreases with the angle of the oxygen jets; however, the transversal velocity, V, increases as a result of the deflection of injectors. The profiles represent well the velocity evolution of the flow near the exit of nozzles. This was verified by two ways: first from the calculation of flow rate by the velocity profiles and second by LDA velocity measurement which the profiles are quite agreed with the PIV results (see Fig. 5a). Table 1 summarizes the initial velocity conditions in the jet axis for both fluids (natural gas and oxygen) as a function of jet angle. It is shown that the turbulence intensity (u'_{cl}/U_{cl}) at the center of jet is 3% for the central jet and between 2 and 3.4% for the side jet. For radial-velocity component, the turbulence intensity is higher.

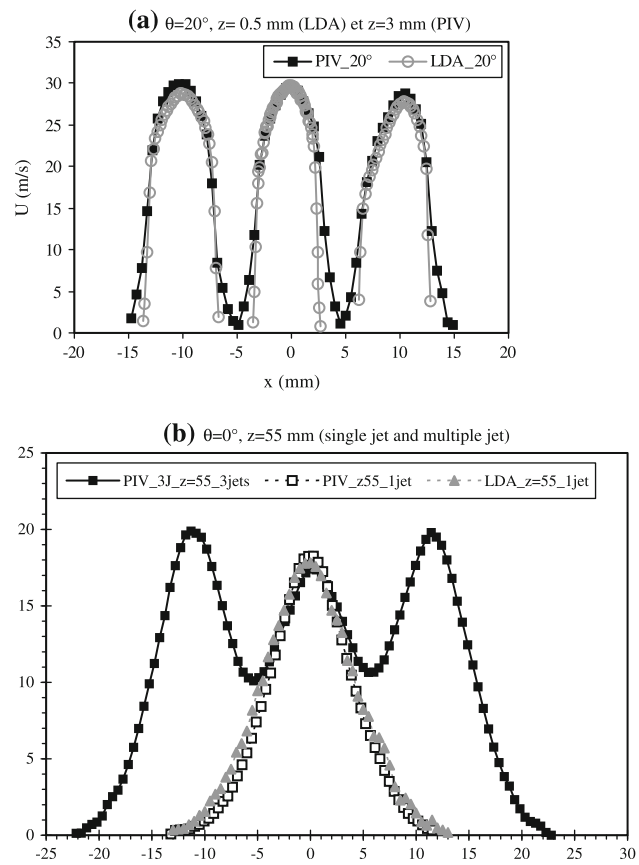

Fig. 5 Radial profiles of axial velocity near the burner in the case of 20°, **a** at $z = 0.5$ et 3 mm, **b** at $z = 55$ mm, PIV–LDA for single jet and PIV for three jet. *Black squares* PIV measurements, *gray symbols* LDA measurements. Setup of LDA measurements: Laser multi-line INNOVA COHERENT, 4 W, IFA 755 Processor, λ: 514.5 nm (*green*) and 488 nm (*blue*)

In order to check the accurate of PIV results, the LDA measurements of some cases are taken. Figure 5 shows a comparison between LDA and PIV measurements in the cases of angles $\theta = 0°$ and $\theta = 20°$. The LDA system consists of a laser multi-line INNOVA COHERENT with 4 W, λ: 514.5 nm (green) and 488 nm (blue), and a IFA

Table 1 Initial conditions of velocities in axis of the jets (subscript cl: value on axis)

	Natural gas jet ($d_{ng} = 6$ mm, $U^0_{ng} = 23.7$ m/s without control)		Oxygen jet ($d_{ox} = 6$ mm, $U^0_{ox} = 25.7$ m/s without control)			
	U_{cl} (m/s)	u'_{cl}/U_{cl} (%)	U_{cl} (m/s)	u'_{cl}/U_{cl} (%)	V_{cl} (m/s)	v'_{cl}/V_{cl} (%)
$\theta = 0°$	30.07	3	32	2.97	0	–
$\theta = 5°$	30.07	3	32	2.7	3.1	25.8
$\theta = 10°$	30.07	3	30.8	2.05	5.9	8.1
$\theta = 15°$	30.07	3	29.9	2.3	7.5	10.6
$\theta = 20°$	30.07	3	29.3	3.4	10.6	10.4
$\theta = 30°$	30.07	3	25.9	2.7	14.6	4.8

755 processor. Figure 5a illustrates radial profiles of the longitudinal velocity near the burner for the injection angle $\theta = 20°$. Globally, a good agreement is found with the both methods. A slight difference is noted at the base of profile which is a priori due to the height of measurements by the two systems. Indeed, the LDA measurements were performed at the exit of nozzles ($z = 0.5$ mm), whereas PIV measurements were performed at the height $z = 3$ mm, it's because of that the velocity profile of PIV is expanded near the wall. It is clear that along the flow, the jets expand because to the interaction with the gas environment. However, at the same position in the flow, the results are very consistent as shown in the Fig. 5b for the single jet and multiple jets. From the velocity profiles of PIV measurements, we calculated the flow rates at the exit nozzles and are in very good agreement with the flow rate imposed. This shows that the PIV measurements therefore can be considered reasonably accurate and can be used to quantitate the interaction of the three jets and the effect of the injection angle.

3.2 Velocity fields

Figure 6 shows an example of instantaneous velocity fields achieved by PIV (taken among 400 instantaneous fields) for oxygen jet angles 0°, 10°, and 20°, with the longitudinal velocity in color scale. The highly three-dimensional flow is observable especially when the jets are inclined, since the flow is characterized by a discontinuous aspect of streamlines. For the central jet, the axial velocity is roughly constant, and the radial velocity is nearly zero in the near nozzle field. More downstream, the side jets affect the central jet, and its radial velocity is no longer zero. Regarding the side jets, except in the case $\theta = 0°$, an incline of velocity vectors is observed, due to the deflection of injectors, and then the impact of these vectors in the internal mixing layers with the central jet. In the far field, vortices appeared in the region of the mixing layer between the jet and the ambient air (for $\theta = 20°$). This is clearly a two-dimensional representation of a three-dimensional phenomenon. At the merging zone of three the jets, some discontinuities in the velocity values are noticed characterizing the three-dimensionality, particularly when the side jets are inclined. This is due to the transverse flow of jets and the elliptical shape of inclined nozzles, as it was shown in the paper of Gutmark and Grinstein (1999) where the entrainment rate and the mixing for elliptic jets are more significant compared to the round jets.

Mean velocity fields in non-reacting flow for the jet angles 0°, 10°, 20°, and 30° are illustrated in Fig. 7. These

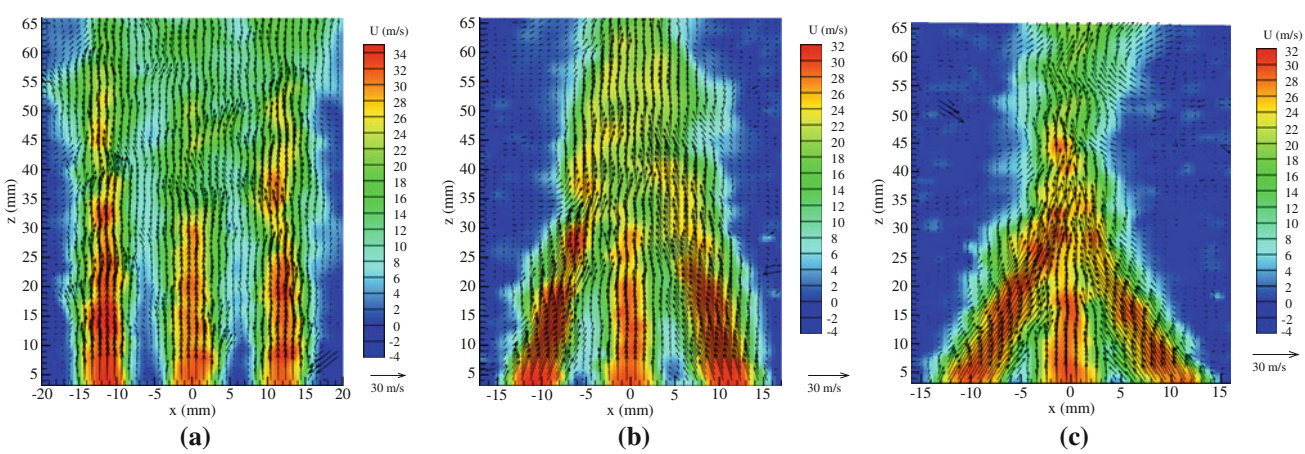

Fig. 6 Instantaneous velocity fields for **a** $\theta = 0°$, **b** $\theta = 10°$, and **c** $\theta = 20°$ in non-reacting flow

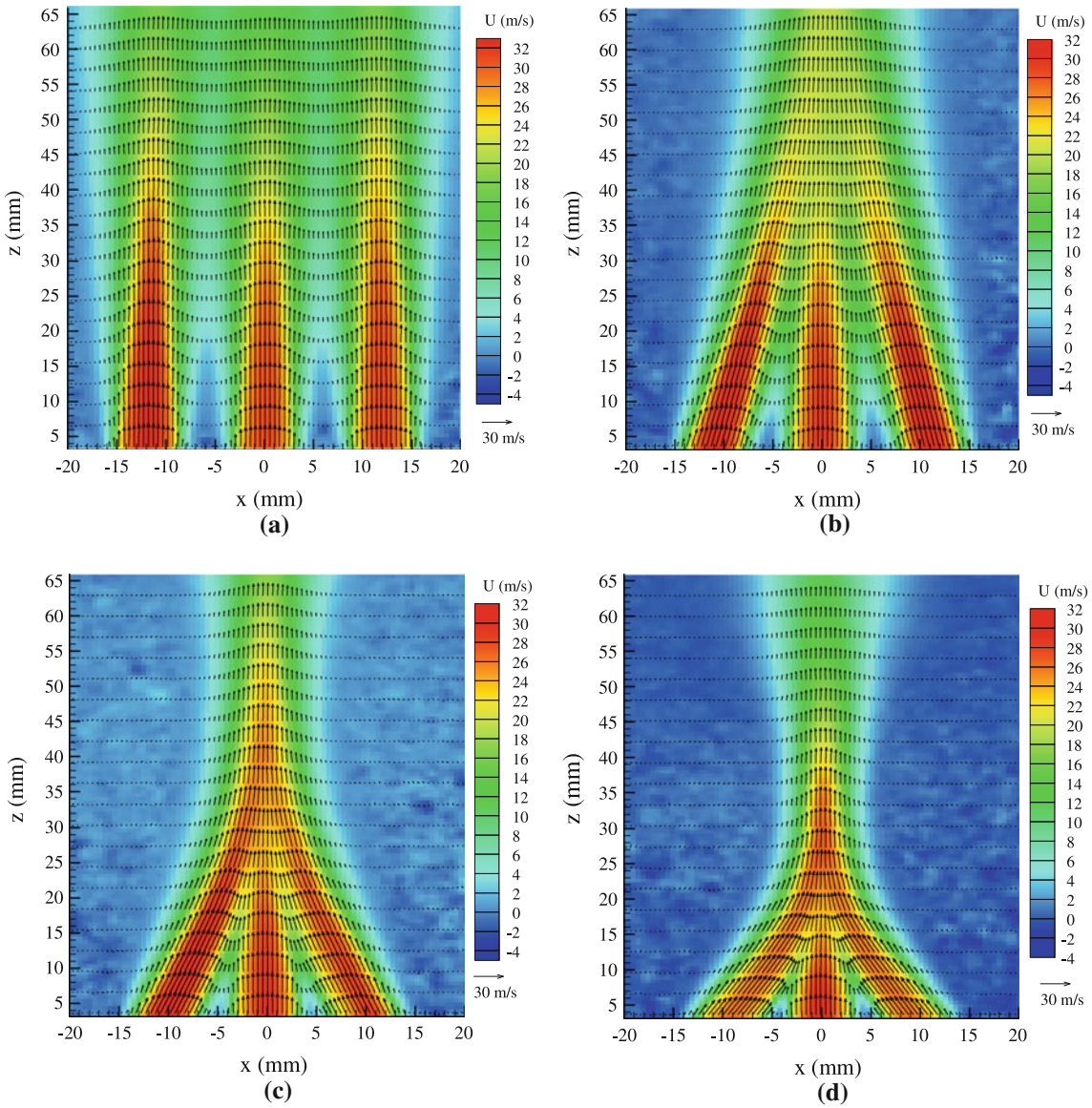

Fig. 7 Mean velocity fields for **a** $\theta = 0°$, **b** $\theta = 10°$, **c** $\theta = 20°$, and **d** $\theta = 30°$ (with longitudinal velocity in color scale) in non-reacting flow

results show that increasing jet angle leads to a decrease of longitudinal velocity and an increase in transversal velocity for the side jets. From the initial state where $\theta = 0°$ to inclined states, the structure of the dynamic field changes. As observed in the figure, the mixing of jets is more upstream for the inclined jet configurations. The merging region starts at 15-mm-high for the straight jets ($\theta = 0°$), and more upstream for inclined jets, 7 mm for $\theta = 10°$ and nearby to the burner for 20° and 30° (at about $z = 3$ mm). The combined region, where the velocity profiles combine to form one profile like a single jet, also starts more and more upstream with the jet angle. An increase in velocities between the jets with increasing jet angle is noted. At the height $z = 10$ mm, the axial velocity is zero between the jets for $\theta = 0°$, whereas it is around 12 m/s for $\theta = 20°$.

3.3 Radial profiles of mean velocities and fluctuations

Figure 8 shows the radial profiles of the mean longitudinal velocity (U) at different heights from the burner for the configurations $\theta = 0°$, 10°, 20°, and 30°. For the straight jets, a classical behavior of multiple jets is found for the distribution of longitudinal velocity, maxima in the center of jets and minima between the jets. In the near burner region ($z = 15$ mm), the distribution of velocity shows maxima and minima corresponding to the three jets, and that the maximum velocity decreases when the jet angle increases. More downstream, when inclining the jets, the extreme velocities begin to disappear into a single maximum located along the axis of the center jet. This combined zone of jets, characterizing a single jet, is reached

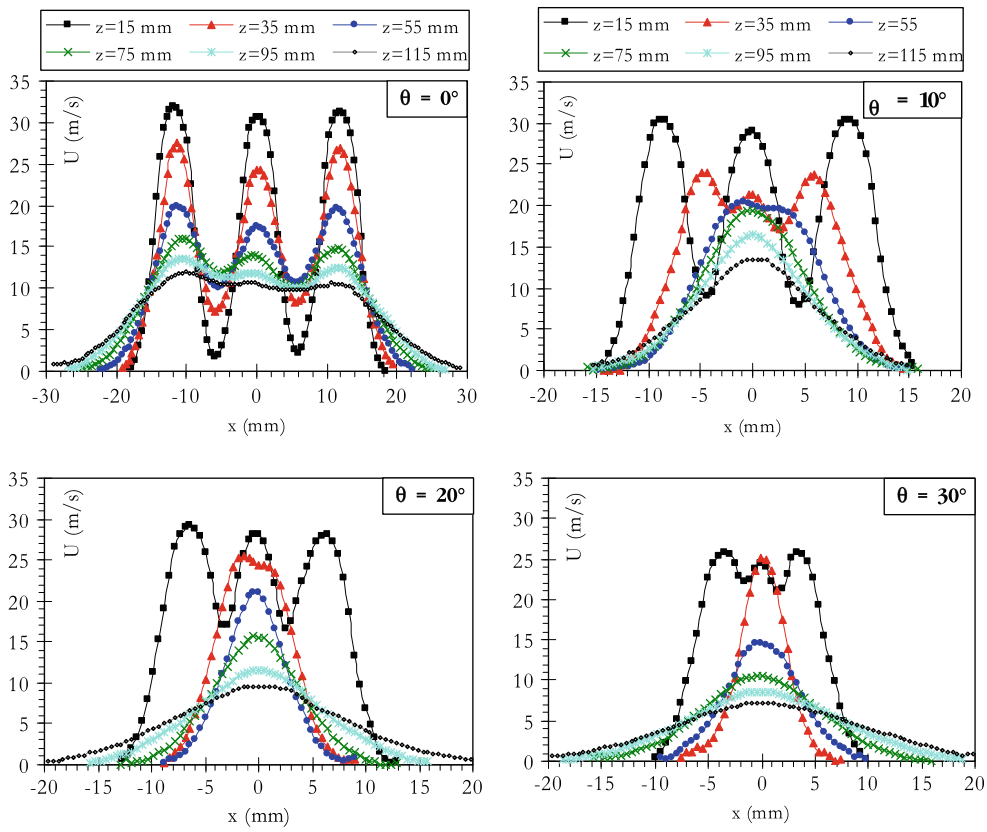

Fig. 8 Radial profiles of mean longitudinal velocity for jet angles 0°, 10°, 20°, and 30° in non-reacting flow at different positions from the burner

earlier when the jet angle increases. Without control ($\theta = 0°$), the combined zone is not reached even at $z = 115$ mm, while for $\theta = 30°$, it is already occurred at $z = 35$ mm. From one configuration to another, it is noticed that the passing from a minimum–maximum profile to that one of a single maximum induces an acceleration of velocity in the axis of the flow. In fact, at $z = 75$ mm, from $\theta = 0°$ to 10°, the centerline longitudinal velocity (U) increases from around 15 to 19 m/s. However, once the combined region is reached, the velocity decreases with the angle of jet injection. The expansion of the flow decreases with the angle in the region near to the burner, and then increases downstream the flow.

The transversal velocity profiles are shown in Fig. 9 at different positions from the burner in non-reacting flow. In the case of straight jets ($\theta = 0°$), the transversal velocity is low and ranges from -1 to 1 m/s, but its maxima increase for $z = 25$ mm and 35 mm and then decrease after $z = 55$ mm. The deflection of injectors leads to an increase in the transversal velocity of the side jets, particularly near the burner since at $z = 15$ mm, the maximum value of V varies from 0.6 to 11 m/s when θ varies from 0° to 30°. As shown in the figure, in cases of inclined jets, the velocity profile is composed of two parts, one positive and one negative with a passage by zero corresponding, respectively, to the side jets (left and right) and the central jet of natural gas.

As observed in Fig. 10, the profiles of velocity fluctuations indicate a very different behavior according to the jet angle (θ) and the axial position of flow (z). In the case without control ($\theta = 0°$), six peaks of fluctuations on the order of 5.5 m/s near the burner are shown, two at the center corresponding to the mixing layers of the central jet and two at both sides of the central jet corresponding to the mixing layers of the side jets. These fluctuation peaks diminish along the flow with decreasing longitudinal velocity in the combined region of flow. From $z = 75$ mm, there are only two fluctuation peaks corresponding to the characteristics of a single jet. Further downstream, the turbulence peaks disappear with the jet angle, and the maximum of fluctuations is located in the flow center.

In Fig. 11, the radial distribution of the turbulence intensity, u'/U, is shown with the jet angle at different axial positions, $z = 15$, 35, 55 and 75 mm. Results of u'/U highlight the interaction zones between the jets and surrounding air as well as between the jets themselves. In the initial region of flow ($z = 15$ mm and $z = 35$ mm), four zones of high turbulence are noticed. Two take place on the outer of the side jets, which dilute the oxygen jets by ambient air. The other two zones are located between the jets representing the jet mixing. Near the exit injectors, the outer zones of turbulence do not seem to be influenced by the increase in jet angle, while the inner zones decrease in intensity, since the merging region is reached faster by

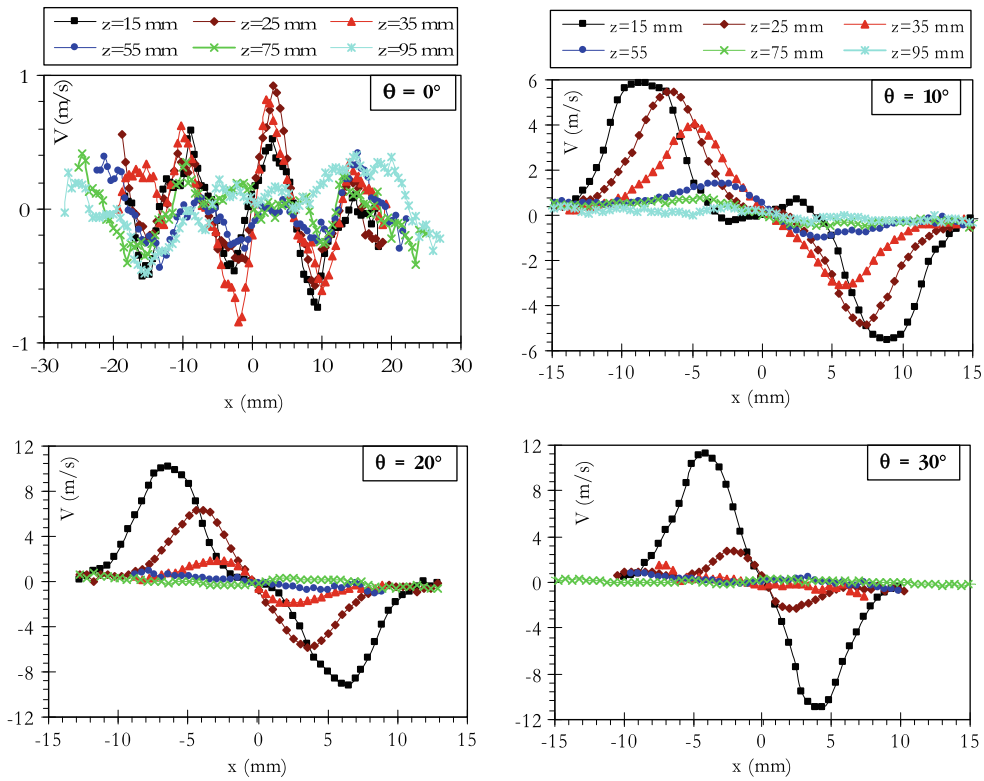

Fig. 9 Radial profiles of mean transversal velocity for jet angles 0°, 10°, 20°, and 30° in non-reacting flow at different positions from the burner

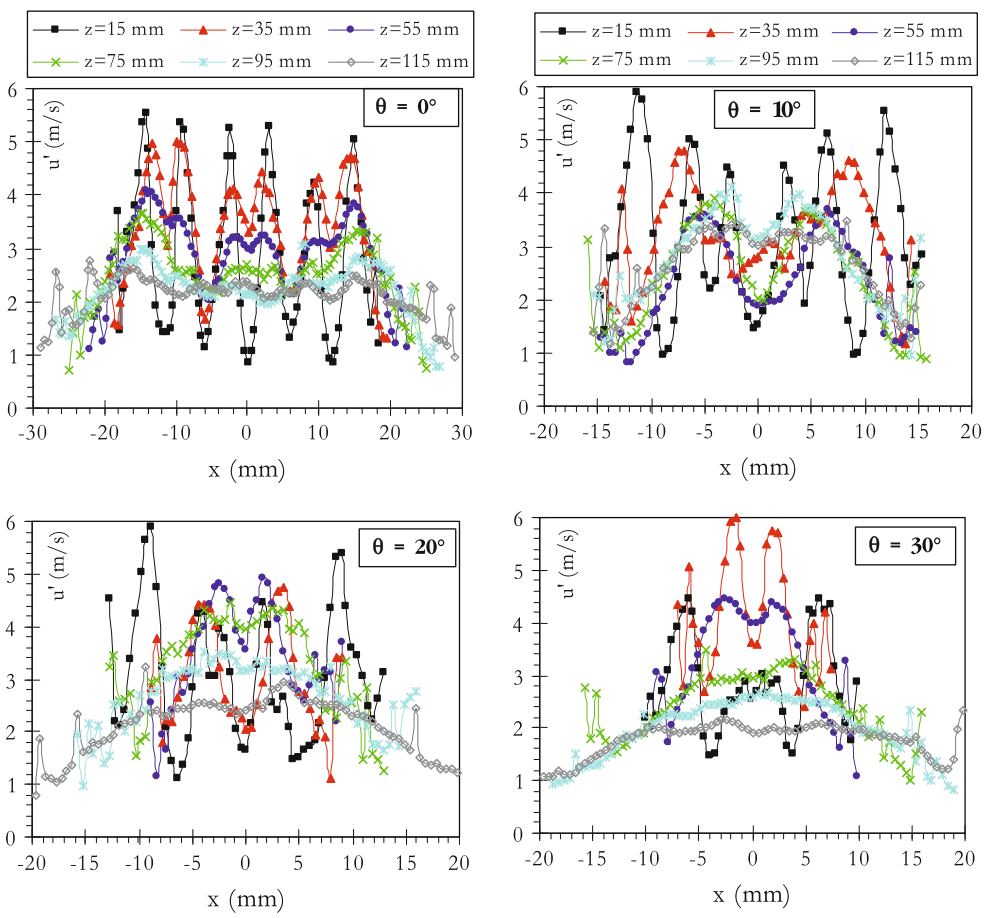

Fig. 10 RMS of longitudinal velocity for jet angles 0°, 10°, 20°, and 30° in non-reacting flow at different positions from the burner

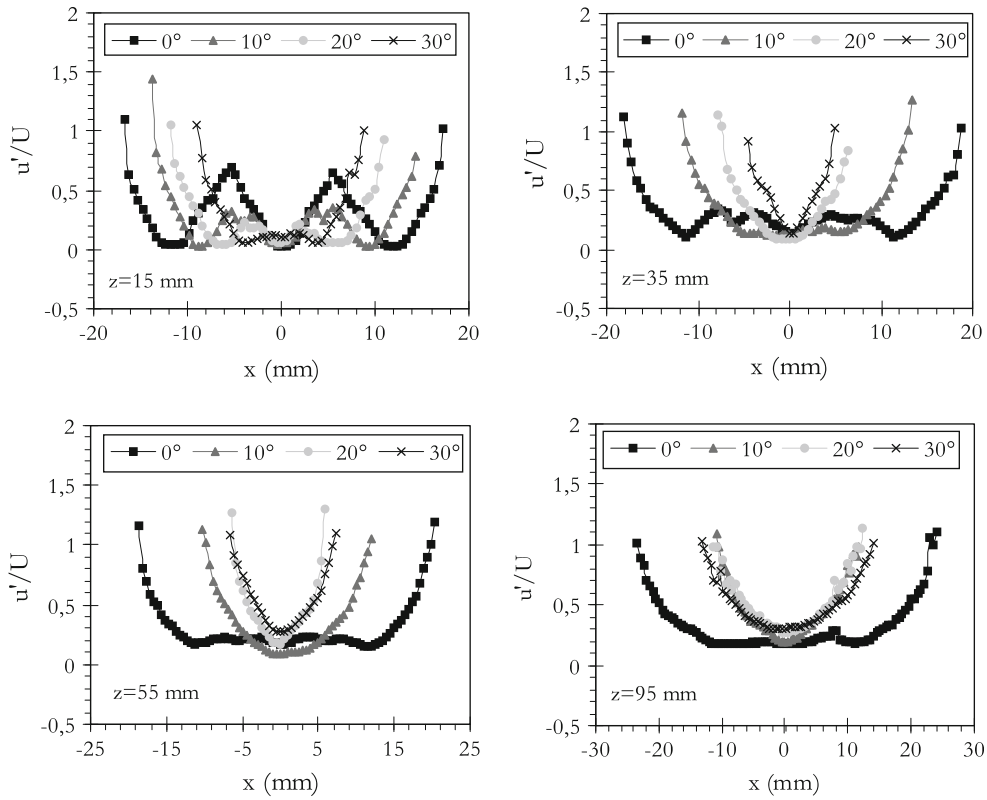

Fig. 11 Turbulence intensity (u'/U) at different axial positions (15, 35, 55, 95 mm) for the jet angles 0°, 10°, 20°, and 30° in non-reacting flow

deflection of jets. Further downstream, when the jets merge, it is found that only the outer zones of turbulence behave as a single jet. At $z = 95$ mm, except for straight jets, it is noted that whatever the jet angle, the intensity profile u'/U is similar owing to the complete merging of jets at this position.

3.4 Velocities along the jet axis

Figure 12 shows the mean velocities along the centerline of the jets (U_{cl} and V_{cl}) for the angles 0°, 10°, 20°, and 30° on the central (NG) and side jets (Ox). The centerline corresponds to the maximum values along the jet. For the central jet, this maximum value locates at the center of nozzle; whereas, for the lateral jet, the maximum value is deviated, because the jet is inclined as shown in Fig. 12d. For the straight jets ($\theta = 0°$), the axial velocity (U_{cl}) follows a classical decrease for the central jet (Fig. 12a) and the side jet (Fig. 12b): first, a very slight decrease to around 26 mm corresponding to the potential core of the jet, then a high decay of U_{cl} which corresponds to the merging of mixing layers, and finally a slow decay up to 65 mm for the natural gas jet and 75 mm for the oxygen jet. Compared to a single jet, the centerline velocity decay in the configuration of three jets is almost the same in the first zone of flow (up to $z = 25$ mm). For greater than 25 mm, the decay is slightly slower for the case of a multiple jet, as shown in Fig. 12a. The same results were found by Moustafa and Rathakrishnan (1993) previously. This may be explained by the results of Moustafa (1994) which found a higher entrainment of the ambient fluid in the case of triple jet configuration. This is true if the exit velocity is almost the same for the three jets, like our configuration. However, if the exit velocity of the central jet is much higher than the one of side jets, for example, the central jet decelerates, and side jets accelerate. Therefore, the centerline velocity decay of the central jet is faster. It is important to note that the velocity decay in multiple jet configurations is highly dependant to the spacing between the jets and exit velocities. Because the difference of velocity between the jets is very low in the present configuration, there is no attraction of jet by other as shown in the configuration studied by Yimer et al. (2001). Tanaka and Nakata (1975) examined the effect of velocity ratio between the jets in a configuration of two-dimensional plane jets. They observed three types of flows according to the velocity ratio: dominant central jet, dominant side jets, and instable state.

When the side jets are inclined ($\theta > 0°$), the behavior of velocity profile changes and U_{cl} varies with the jet angle (Fig. 12). A decrease in the length of potential core and therefore a fast decrease in the longitudinal velocity in the first zone of the flow in particular for the central jet are noted. The decrease in the velocity (U_{cl}) is more important

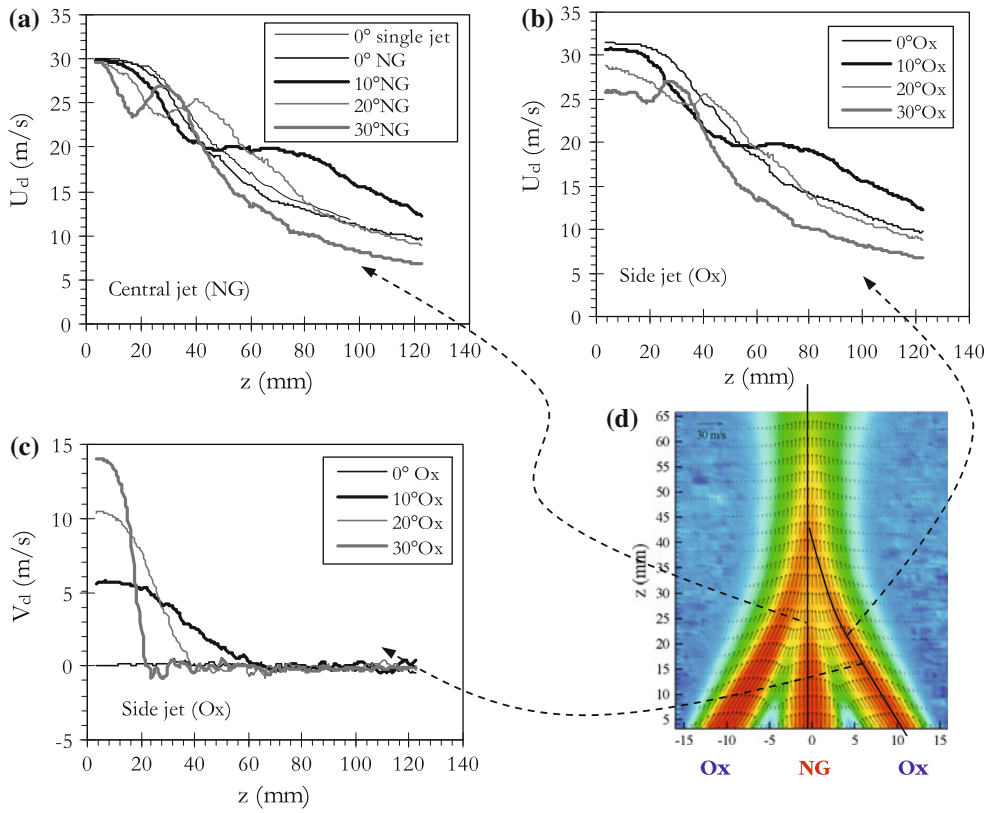

Fig. 12 Mean longitudinal (U_{cl}) and transversal (V_{cl}) velocities along the centerline of the central jet (NG) and the side jet (Ox)—[Non-reacting flow, jet angles: 0°, 10°, 20°, and 30°]

with the jet angle near the burner. After this first part of the flow (Fig. 12a), a plateau for the case $\theta = 10°$ at $z = 48$ mm and an increase in U_{cl} for the cases $\theta = 20°$ at $z = 28$ mm and $\theta = 30°$ at $z = 18$ mm are observed. This behavior is attributed to the merging of three jets leading to an acceleration of flow at the beginning of the combined region. The distribution of U_{cl} for the oxygen jet differs from the one of the natural gas jet in the near field, since it shows a little decrease and then a slight interval for the variation point of velocity along z for the oxygen velocity. Further downstream, the U_{cl} profiles for both fluids are the same, because the jets are combined to form a single jet.

The transversal velocity along the centerline (V_{cl}) is zero for the central jet, but it is not the case for the side jet if they are inclined (14 m/s for $\theta = 30°$ close to the burner) as shown in Fig. 12c. The decay of V_{cl} for the oxygen is faster for an increasing angle of injection. It is noticed that according to the angle of injection, the decay of velocity V_{cl} until zero is located further upstream along the z position, at 60, 40, and 20 mm, respectively, for jet angles 10°, 20°, and 30°.

3.5 Oxy-combustion effects on dynamic field

Figure 13 shows the mean velocity fields in the reacting flow for jet angles 0° and 30°. These vector fields show a significant difference between non-reacting and reacting flow. The first remark concerns the more significant velocities above the stabilization point in the reacting flow. The hot environment (~500°C) and the presence of a reaction zone lead to a fast expansion of gases and a decrease of the entrainment of ambient fluid due to the presence of flame, therefore to an acceleration of the flow. A greater radial expansion in combustion in particularly above the stabilization zone of the flame is observed.

The radial distributions of mean longitudinal velocities in reacting flows are shown in Fig. 14 for the jet angles 0°, 10°, 20°, and 30° at different axial positions. For the straight jets ($\theta = 0°$), the longitudinal velocity (U) decreases along the flow; however, this decrease is less significant compared to non-reacting flow. Indeed, at $z = 75$ mm position, the natural gas and oxygen velocities are, respectively, 22.5 and 27.5 m/s in reacting flow, whereas in non-reacting flow, they are 14 and 15 m/s. For the inclined jets ($\theta > 0°$), the longitudinal velocity in combustion keeps higher values even at more significant heights from the burner. In fact, as indicated, for the case $\theta = 10°$, the maximum velocity in the central jet is about 27 m/s at $z = 15$, 55 and 95 mm which is not the case in non-reacting flow where the velocity decreases with z distance. For a higher angle ($\theta = 20°$), the velocity U is more significant and reaches at 30 m/s at $z = 55$ mm. For $\theta = 30°$, at the positions $z = 35$ and 55 mm, the longitudinal velocity in the reacting flow is twice as high

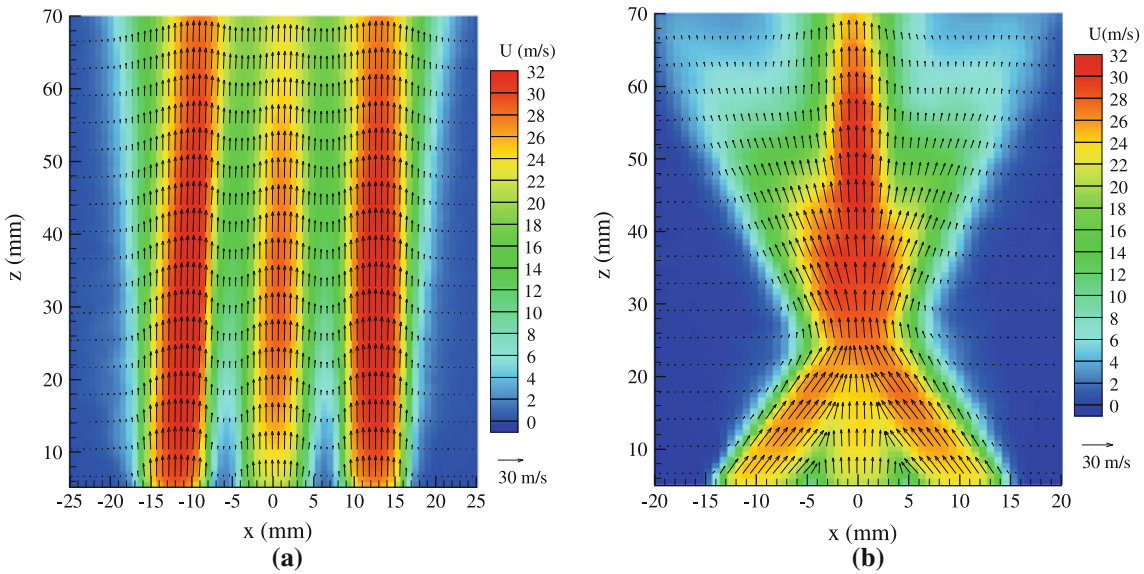

Fig. 13 Mean velocity fields for the oxygen jet angle **a** $\theta = 0°$ and **b** $\theta = 30°$ in reacting flow

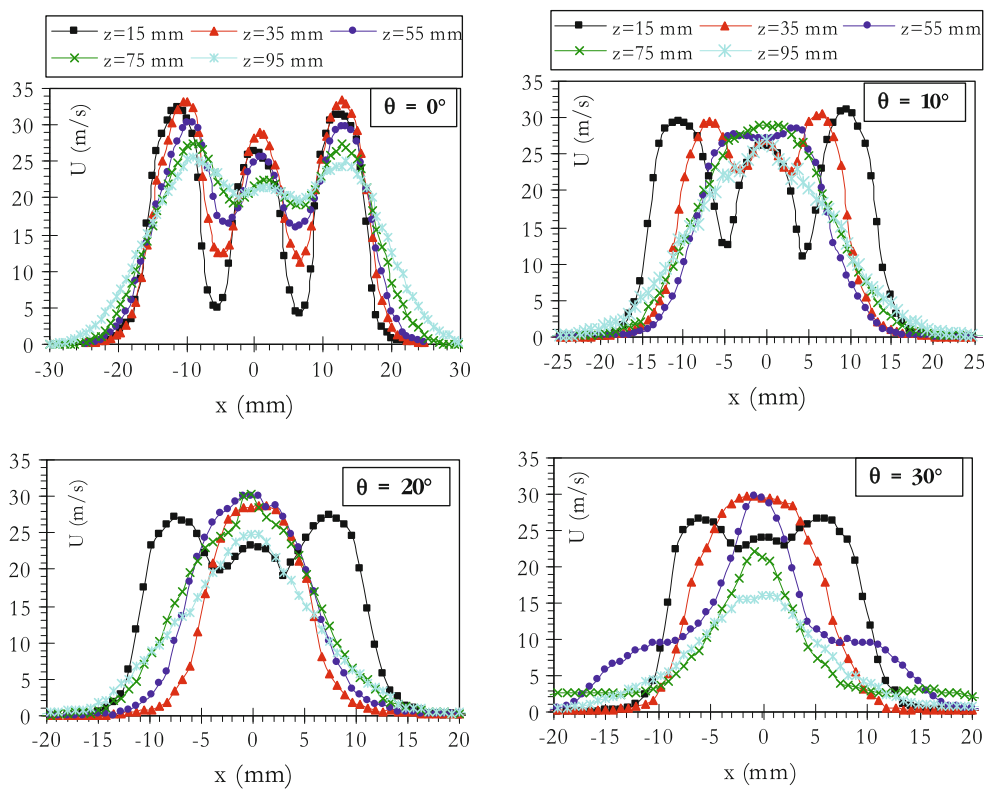

Fig. 14 Radial profiles of mean longitudinal velocity for jet angles 0°, 10°, 20°, and 30° in reacting flow at different positions from the burner

compared to the non-reacting flow (30 and 15 m/s). It is noted that for this configuration ($\theta = 30°$), a greater radial expansion caused by the flame, particularly at $z = 55$ mm location.

The transversal velocity profiles in the reacting flow, presented in Fig. 15, show that the increase in jet angle leads to an increase in velocity in the radial direction. For the jet angle $\theta = 30°$, it is noted that the two parts (right and left) of V reverse the sign at $z = 55$ mm position. This can be explained by the presence of flame front, which influences the flow direction.

The axial distribution of the longitudinal velocities (U_{cl}) in reacting flow is shown in Fig. 16 for the central and oxygen jets. The velocity U_{cl} increases with z distance in the first zone up to a maximum, then it conserves a high value that varies with the jet angle, finally, it slightly

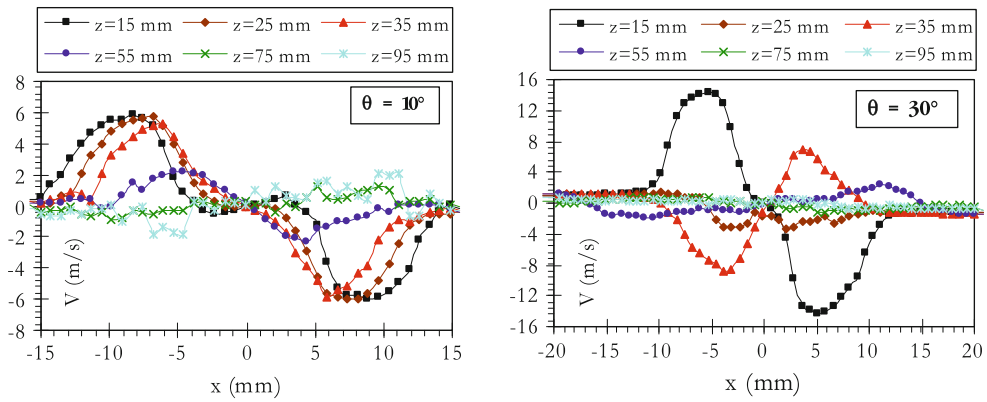

Fig. 15 Radial profiles of mean transversal velocity for jet angles 10° and 30° in reacting flow at different positions from the burner

decreases along the centerline. The oxy-fuel combustion considerably influences the flow throughout the studied domain. In fact, in comparison with the non-reacting flow in the initial region, the flow is slower for both fluids, and then accelerates under the influence of high temperatures of the flame. Starting from $z = 40$ mm, the centerline velocity is higher in the case of the reacting flow for the four configurations of jet angles studied. After the stabilization point, the flow velocity is higher in the reactive case than that in the non-reactive case. This is due both to the rapid expansion of burnt gases which accelerates the flow and a retardation of mixing due to the presence of a flame.

Figure 17 shows the radial profiles of turbulence intensity (u'/U) at heights $z = 15$, 35, 55 and 95 mm for reacting flow. As in reacting flow, in the initial zone of flow, the mixing layers generating turbulence are found, the inner zone at the interface of the natural gas-oxygen and the outer zone at the interface of the oxygen–ambient fluid. Comparison of Figs. 11 and 17 indicates that the turbulence intensity on the jet axis is higher in the non-reactive case. Indeed, at $z = 35$ mm for $\theta = 0°$, $u'_{cl}/U = 0.15$ for the natural gas jet and 0.12 for the oxygen jet in the non-reacting flow; whereas $u'_{cl}/U = 0.08$ for the natural gas jet and 0.05 for the oxygen jet in the reacting flow. This can be explained by the increase in temperature due to heat release from flame, which leads a retardation of turbulence formation at the upstream region of the flow. This obvious result is well reported in the literature such as Takagi et al. (1981). Above the stabilization region, the velocity fluctuations are higher in the case with flame than that without flame. The higher velocities induced by the temperature rise keep a higher turbulence in the reactive case in the downstream of the jet. These results are as well observed by Takagi et al. (1981) on a H_2–N_2/Air flame.

In the inter-jets region, a slight increase in velocity fluctuations in the reactive case from the non-reactive case is observed. Since, at $z = 35$ mm, $u'/U = 0.3$ in non-reacting flow and 0.4 in reacting flow. This increase may be due to the presence of the lift-off point of flame which is located spatially between the jets and is greatly fluctuated.

Downstream of the flow, as the height increases, the two inner mixing layers disappear, since the jets merge; whereas the outer zones of turbulence remain, and the turbulence intensity u'/U keeps higher values.

After describing the dynamic field of the total flow, it seems interesting to establish the velocity values obtained at the stabilization points of the studied oxy-fuel flames. The mean longitudinal velocities at the lift-off positions of

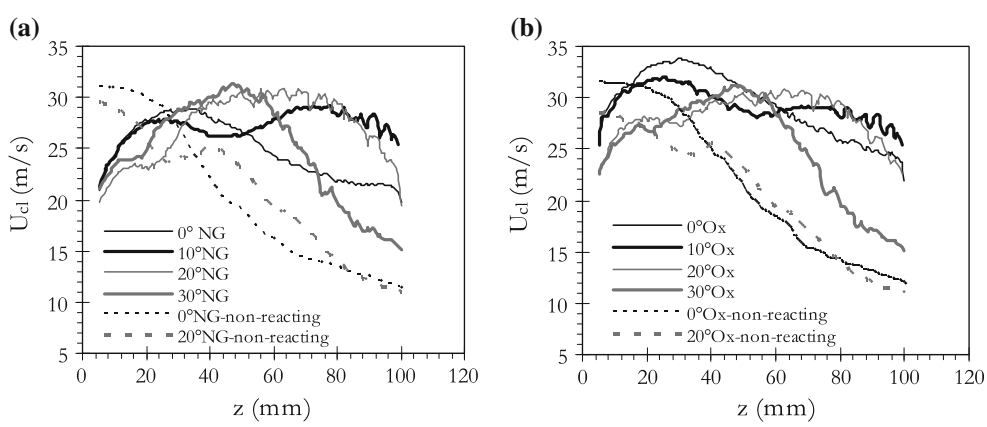

Fig. 16 Mean longitudinal velocity (U_{cl}) along the centerline of the central jet (**a**) and the side jet (**b**)—[Reacting flow, angles: 0°, 10°, 20°, and 30°, and non-reacting flow for 0 and 20°]

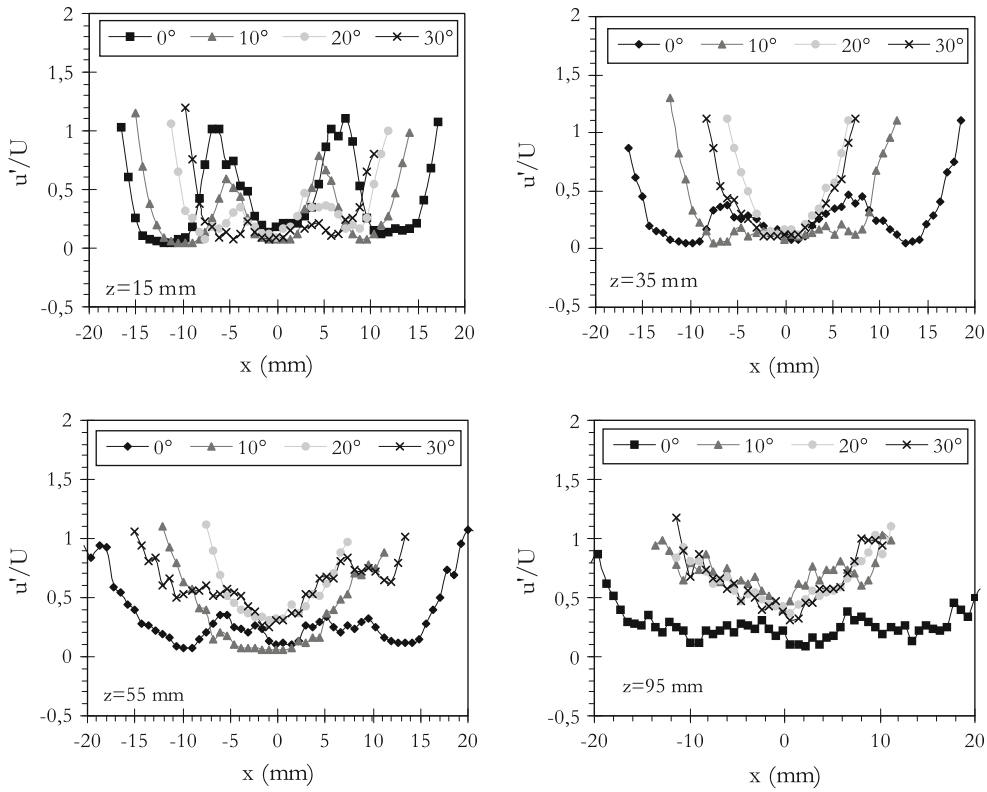

Fig. 17 Turbulence intensity (u'/U) at different axial positions (15, 35, 55, 95 mm) for the jet angles 0°, 10°, 20°, and 30° in reacting flow

flame are summarized in Table 2. The lift-off positions (H_{lo} and X_{lo}) were measured by the OH chemiluminescence in a previous work (Boushaki et al. 2007). The stabilization point is located between the jets where the local velocity has a minimum. The velocity (U) obtained neighbor the stabilization point in the four cases (9.68–18 m/s) is higher than the laminar flame velocity ($S_L = 3.9$ m/s) for a natural gas-oxygen flame. In previous studies for a jet flame, it was found that the mean axial velocity measured at the flame base is around 1.2 S_L (0.52 m/s) (Schefer and Goix 1998), 2 S_L (Muniz and Mungal 1997), 2.7 S_L (Watson et al. 1999). This deviation is due to the geometry of the separated jet burner where the distance between the nozzles is a determining factor for the stabilization flame, also when pure oxygen is used as oxidizer.

Table 2 Mean longitudinal velocity at the lift-off height of flame

Jet angle θ (°)	Vertical position H_{lo} (mm)	Horizontal position X_{lo} (mm)	Velocity at stabilization point, U_{lo} (m/s)
0	27	6.15	9.68
10	19.21	4.78	15.48
20	12.23	4.13	18
30	6.55	3.8	14.38

4 Conclusions

An experimental study of development and structure of flow from an oxy-fuel burner with separated jets was investigated in reacting and non-reacting flows. Measurements by PIV technique enabled the characterization of the behavior of jets and the distribution of velocity fields.

Results show that the inclination of side jets toward the central jet improves the mixing and thus accelerates the merging, and then the combining of jets where velocity profiles become uniform to form a single jet profile. The deflection of injectors induces a transversal velocity in the first zone of flow increasing with the jet angle. Along the flow, this velocity decreases, since the side jets impact the central jet, and the flow forms a single jet. The axial velocity along the centerline of jets (U_{cl}) follows an ordinary evolution for the straight jets ($\theta = 0°$), whereas it changes behavior with the deflection of jets ($\theta > 0°$). Indeed, the potential core length decreases, and therefore the longitudinal velocity decreases fastly in the first zone of the flow with the jet angle, in particular for the central jet. After the first region, the U_{cl} profile presents a plateau (for 10°) and an increase in velocity (for 20° and 30°) due to the union of jets leading a slight acceleration of flow at the beginning of the combined zone of jets. The results for the turbulence intensity highlighted the zones of high turbulence in the outer (ambient air-oxygen) and in the inner (oxygen-natural gas) mixing layers. The inner zones of

turbulence decrease in intensity along the axis of the burner and then disappear in the combined region of jets.

The oxy-fuel combustion generates a significant change on the distribution of dynamic fields. The velocities are higher (sometimes twice as high) compared to non-reacting flow, in particular above the stabilization region. This is due both to the rapid expansion of burnt gases which accelerates the flow and a retardation of mixing due to the presence of a flame. Also, a greater radial expansion in reacting flow particularly above the lift-off position of flame was observed. Near the burner, the intensity turbulence is lower in reacting flow, but it higher downstream in consequence of the high velocities due to the presence of flame.

References

Baukal CE (2003) Industrial burners handbook. CRC Press, Boca Raton

Baukal CE, Gebhart B (1997) Oxygen-enhanced/natural gas flame radiation. Int J Heat Mass Transf 40(11):2539–2547

Boushaki T, Sautet JC, Salentey L, Labegorre B (2007) The behaviour of lifted oxy-fuel flames in burners with separated jets. Int Commun Heat Mass Transf 34:8–18

Boushaki T, Mergheni MA, Sautet JC, Labegorre B (2008) Effects of inclined jets on turbulent oxy-flame characteristics in a triple jet burner. Exp Therm Fluid Sci 32:1363–1370

GEFGN (1979) Utilisation de l'oxygène dans les flammes de diffusion de gaz naturel. Revue Générale de Thermique (214):643–649

Gutmark EJ, Grinstein FF (1999) Flow control with non circular jets. Annu Rev Fluid Mech 31:239–272

Ivernel A, Marque D (1975) Flammes industrielles de gaz naturel. Influence de la suroxygénation du comburant. In: 2nd European combustion symposium, pp 121–128

Krothapalli A, Bagadanoff D, Karamchetti K (1980) Development and structure of a rectangular jet in a multiple jet configuration. AIAA J 18(8):945–950

Leite AOP, Ferreira MA, Carvalho JA (1996) An investigation of multiple jet acetylene flames. Int Commun Heat Mass Transf 23(7):959–970

Lenze B, Milano ME, Günther R (1975) The mutual influence of multiple jet diffusion flames. Combust Sci Tech 11:1–8

Menon R, Gollahalli SR (1988) Combustion characteristics of interacting multiple jets in cross flow. Combust Sci Tech 60:375–389

Moawad Ahmed AK, Rajaratnam N, Stanley SJ (2001) Mixing with multiple circular turbulent jets. J Hydraul Res 39(2):163–168

Moustafa GH (1994) Main features of overexpanded triple jets. AIAA J 32(11):2205–2209

Moustafa GH, Rathakrishnan E (1993) Studies on the flow field of multiple jet with square configuration. AIAA J 31(7):1189–1190

Muniz L, Mungal MG (1997) Instantaneous flame stabilization velocities in lifted jet diffusion flames. Combust Flame 11:16–31

Pani B, Dash R (1983) Three dimensional single and multiple jets. J Hydraul Eng 109(2):254–269

Perthuis E (1983) La combustion industrielle. Editions Technip, Paris

Raghunathan S, Reid IM (1981) A study of multiple jet. AIAA J 19:124–127

Sautet JC, Boushaki T, Salentey L, Labegorre B (2006) Oxy-combustion properties of interacting separated jets. Combust Sci Tech 178:2075–2096

Schefer RW, Goix PJ (1998) Mechanism of flame stabilization in turbulent lifted jet flames. Combust Flame 112:559–574

Simonich JC (1986) Isolated and interacting round parallel heated jets. AIAA Pap 86–0281

Takagi T, Shin HD, Ishio A (1981) Properties of turbulence in turbulent diffusion flames. Combust Flame 40:121–140

Tanaka E (1974) The interference of two dimensional plane jet (2rd report, experiments on the combined flow of dual jet). Bull JSME 17(109):920–927

Tanaka E, Nakata S (1975) The interference of two dimensional plane jet (3rd report, the region near the nozzle in triple jets). Bull JSME 18(124):1134–1141

Watson KA, Lyons KM, Donbar JM, Carter CD (1999) Scalar and velocity field measurements in a lifted CH_4-air diffusion flame. Combust Flame 117:257–271

Yimer I, Becker HA, Grandmaison EW (1996) Development of flow from multiple jet burner. AICHE J 74:840–851

Yimer I, Becker HA, Grandmaison EW (2001) The strong jet/weak jet problem: new experiments and CFD. Combust Flame 124:481–502